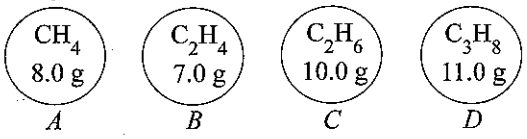
31. The power of an electric heater is 1 000 W. The amount of electrical energy spent when this heater operated for a period of 2 hours was,
(1) 0.5 kW h. (2) 2 kW h. (3) 500 kW h. (4) 2 000 kW h.

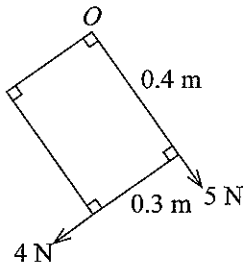
32. Which of the following metals can be used for cathodic protection of iron?
(1) Ag (2) Cu (3) Sn (4) Zn

33. The force exerted vertically upwards on a rocket of mass m due to combustion of fuel is R. If g is the acceleration due to gravity, the acceleration with which the rocket moves is,
(1) $\frac{R-mg}{m}$. (2) $\frac{R}{m}$. (3) $\frac{R-m}{mg}$. (4) $\frac{R}{mg}$.

34. Given in the figure are masses of four gaseous hydrocarbons. In which of them equal number of molecules are contained? (C=12, H = 1)
(1) *A* and *B* (2) *A* and *C*
(3) *B* and *D* (4) *C* and *D*

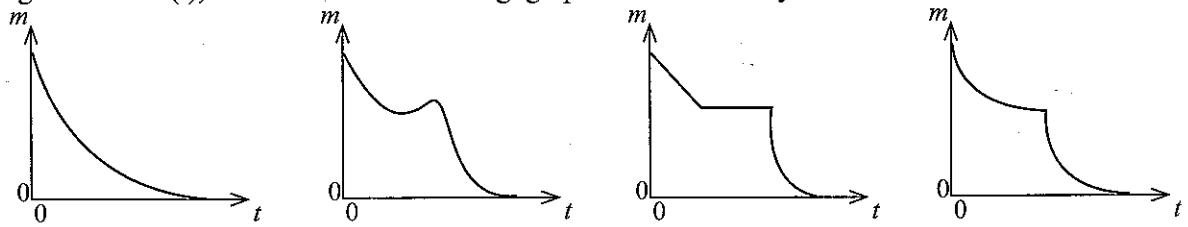


35. A light, rectangular lamina 0.4 m × 0.3 m is pivoted at point *O* and two forces 5 N and 4 N are applied on it as shown in the diagram. What is the moment of force on the lamina around the point *O*?
(1) 1.6 N m anticlockwise (2) 1.6 N m clockwise
(3) 1.5 N m anticlockwise (4) 1.5 N m clockwise



36. The boiling points of hydrogen chloride and water are -85 °C and 100 °C respectively. Which of the following statements explains the cause for this difference?
(1) The intermolecular forces among hydrogen chloride molecules are weaker than the intermolecular forces among water molecules.
(2) The relative molecular mass of hydrogen chloride is greater than the relative molecular mass of water.
(3) The covalent bonds among the atoms are stronger in water than those in hydrogen chloride.
(4) Water molecules are polar whereas hydrogen chloride molecules are non-polar.

37. A sheet of zinc suspended on a spring balance is immersed in a hydrochloride acid solution bringing them to react. Some time after the beginning of the reaction another quantity of the acid solution was added into the reaction mixture. If the spring balance reading (m) was plotted against time (t), which of the following graphs is most likely to be obtained?



(1) (2) (3) (4)

38. Of the following, which one is the **false** statement?
(1) Acid rains occur due to release of chlorofluorocarbons to the atmosphere.
(2) Greenhouse effect is necessary to maintain life.
(3) Greenhouse effect affects global warming.
(4) Oxides of nitrogen affect depletion of the ozone layer.

39. Which of the following disaster conditions has the minimum influence of humans?
(1) occurrence of tsunami (2) occurrence of photochemical smog
(3) spread of invasive alien plants (4) invasion of settlements by wild animals

40. In a traditional tank (wewa), what is the structure that prevents the erosion of the tank bund?
(1) bisokotuwa (2) ralapanawa (3) out spill (4) wew thaulla

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